

Dec-25-0649

MCA-7303 (Cloud Computing)

MCA-3rd NEP

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note: Attempt five questions in all, select one question each from section A, B, C, D. Section E (Question-9) is compulsory.

SECTION-A

1. (a) Explain the NIST-defined essential characteristics of Cloud Computing and analyze how each characteristic impacts cloud service design. (6)
(b) Explain the role and importance of the DMTF Cloud Management Working Group in defining cloud standards and reference architectures. (6)
2. (a) Discuss the architectural design and components of IBM Cloud Computing Reference Architecture and explain how it supports workload deployment and service management. (6)
(b) How do multi-tenancy, service abstraction, and dynamic provisioning affect cloud performance and security? Discuss. (6)

SECTION-B

3. (a) Explain and differentiate between IaaS, PaaS, and SaaS service models with suitable technical examples. (6)
(b) Define multi-tenancy in cloud computing. Discuss its architectural design and security implications in shared (6)

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4. (a) What are the different types of virtualization, and how does each of them enhance cloud efficiency? Explain in detail. (6)
(b) Explain the integration of mobile computing with cloud services. How does mobile cloud computing address the limitations of mobile devices? (6)

SECTION-C

5. (a) Discuss the architectural components of a Cloud Database. Explain how scalability, replication, and consistency are achieved in cloud-based database systems. (6)
(b) Compare the Google File System and Hadoop Distributed File System in terms of architecture, fault tolerance, and metadata management. (6)
6. (a) Describe the architecture and data model of Google Bigtable. How does it influence the design of modern NoSQL systems? (6)
(b) Discuss the structure and functionality of MongoDB. How does it achieve schema flexibility and horizontal scalability in cloud environments? (6)

SECTION-D

7. (a) Discuss the key security challenges in cloud computing arising from multi-tenancy, virtualization, and data outsourcing. (6)
(b) Discuss infrastructure security issues in cloud data centers. How can network segmentation and virtual firewalls mitigate these risks? (6)
8. (a) Explain the role of hypervisor security in virtualized cloud infrastructures. Discuss potential attack surfaces and countermeasures. (6)

- (b) How does cloud security differ in public, private, and hybrid deployments? Discuss the security implications of each model. (6)

SECTION-E (COMPULSORY QUESTION)

- (a) How does cloud elasticity differ from scalability in performance management?
- (b) How does risk management in cloud differ from traditional IT infrastructure?
- (c) What are the key layers of the Cloud Computing Reference Model?
- (d) Why is workload portability important in hybrid cloud deployments?
- (e) Distinguish between Type 1 and Type 2 hypervisors.
- (f) How are Service-Oriented Architecture principles integrated into cloud service models? (6×2=12)